

Classwork – (18-09-2026)

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- **Local** : Calculate NDVI, NDWI, NDBI
 - Waterbody detection: NDWI thresholding
 - Calculate the total number of waterbodies and area
 - Expected outcomes:
 - Raster data representing NDVI, NDWI, and NDBI
 - Vector map of all waterbodies with attribute table having info on area, perimeter, total count and total area
- **Focal**: Moving window
 - Select a specific band which shows sharp features over Hyd region and smooth it out with varying neighborhood size 3*3, 5*5
 - Observe original band and processed rasters
 - Expected outcomes:
 - Snapshot showing regions over Hyd. In original band, 3*3 focal op., 5*5 focal op.

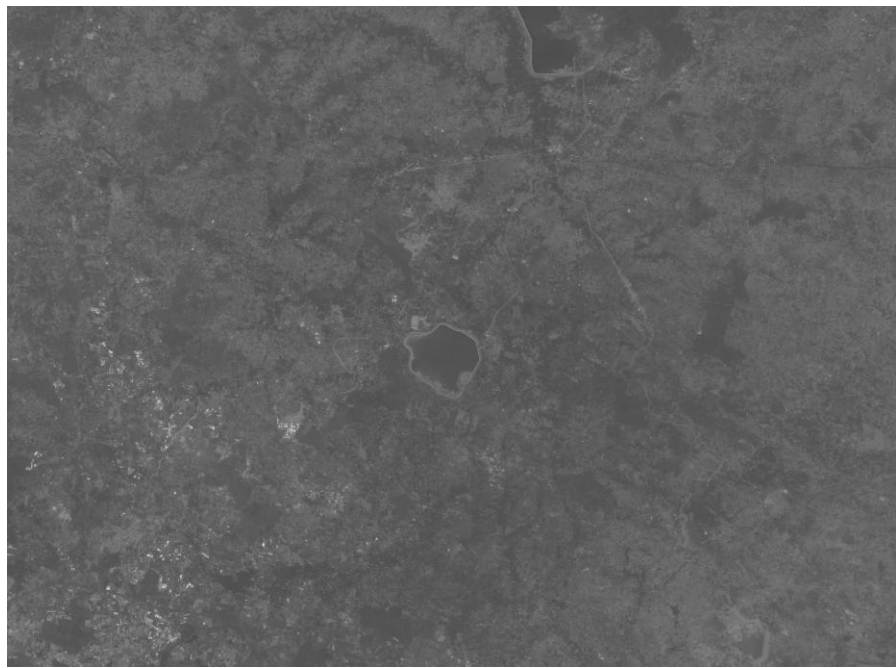
LISS-III Spectral Bands and Wavelengths

- Green (B2): 0.52 - 0.59 μm
- Red (B3): 0.62 - 0.68 μm
- NIR (B4): 0.77 - 0.86 μm
- SWIR (B5): 1.55 - 1.70 μm

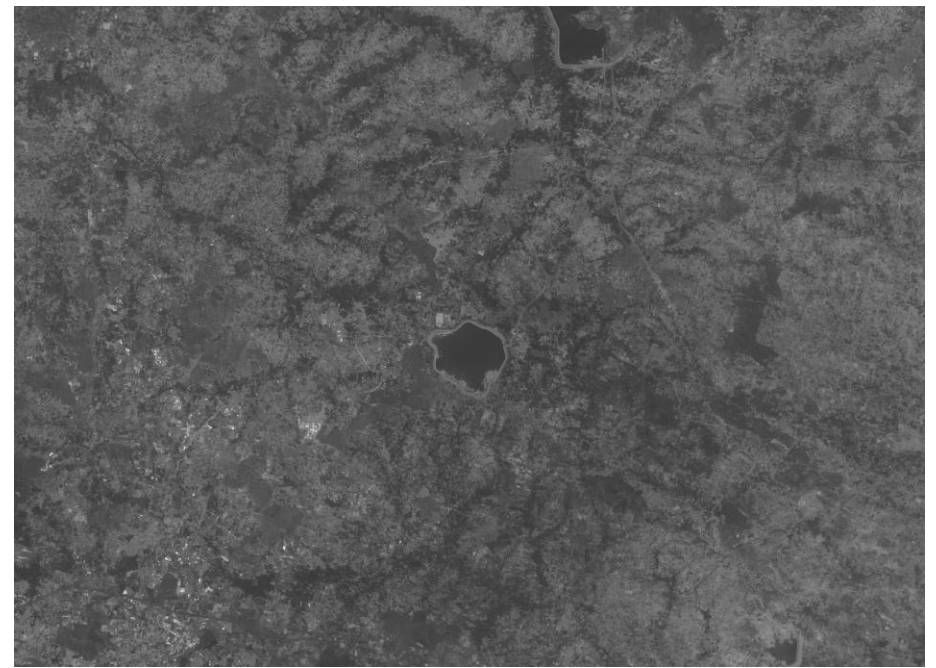
$$NDVI = \frac{NIR - RED}{NIR + RED}$$

$$NDWI = \frac{GREEN - NIR}{GREEN + NIR}$$

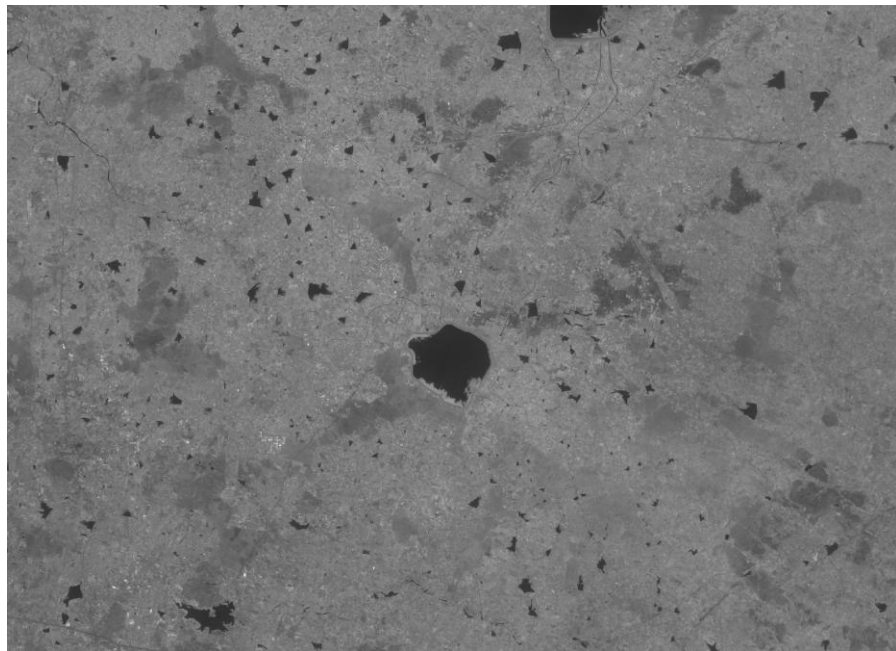
$$NDBI = \frac{SWIR - NIR}{SWIR + NIR}$$



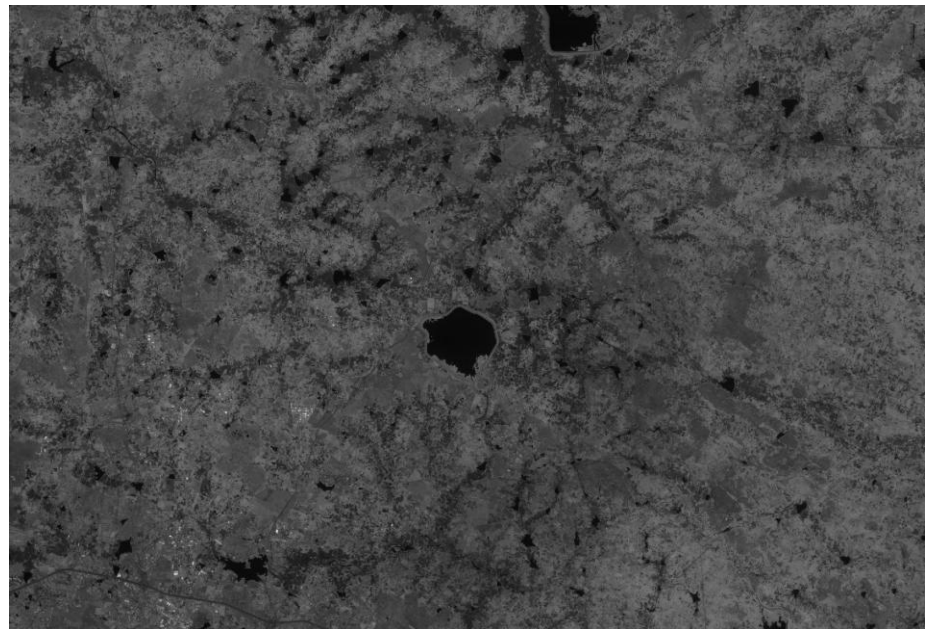
B2



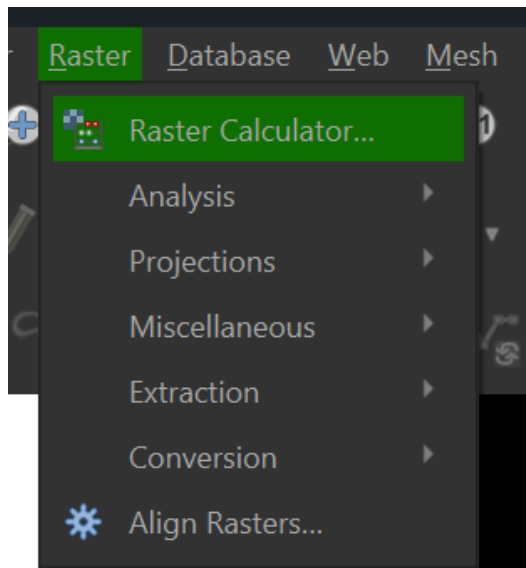
B3



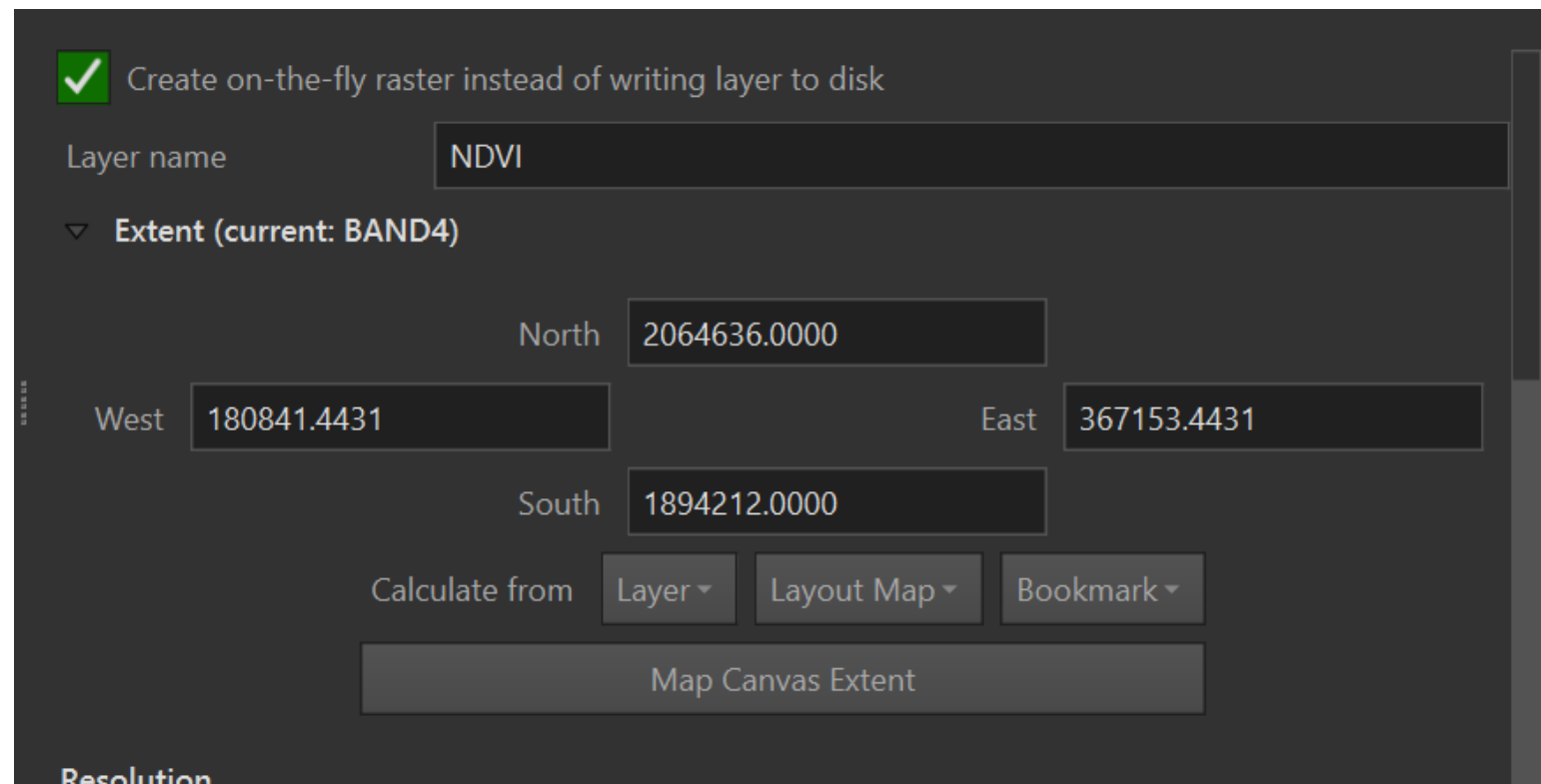
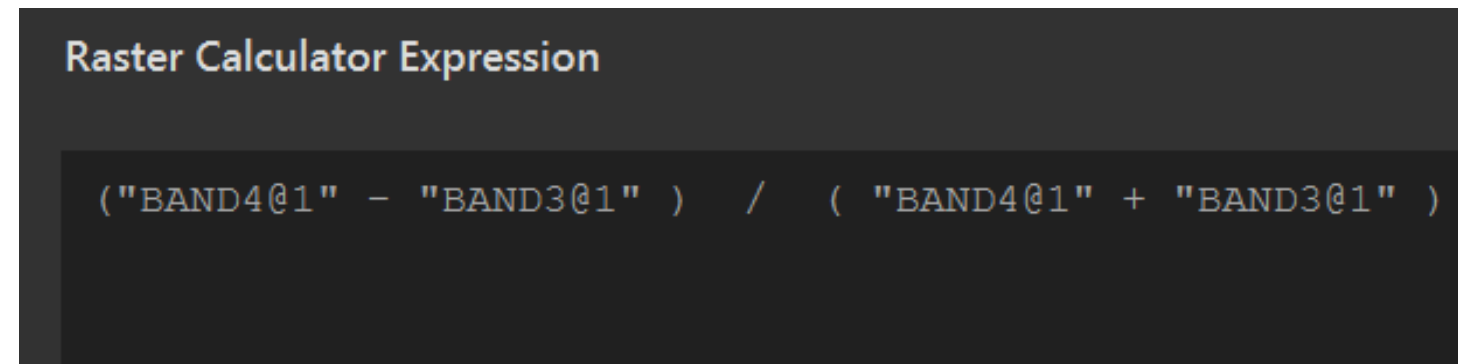
B4



B5

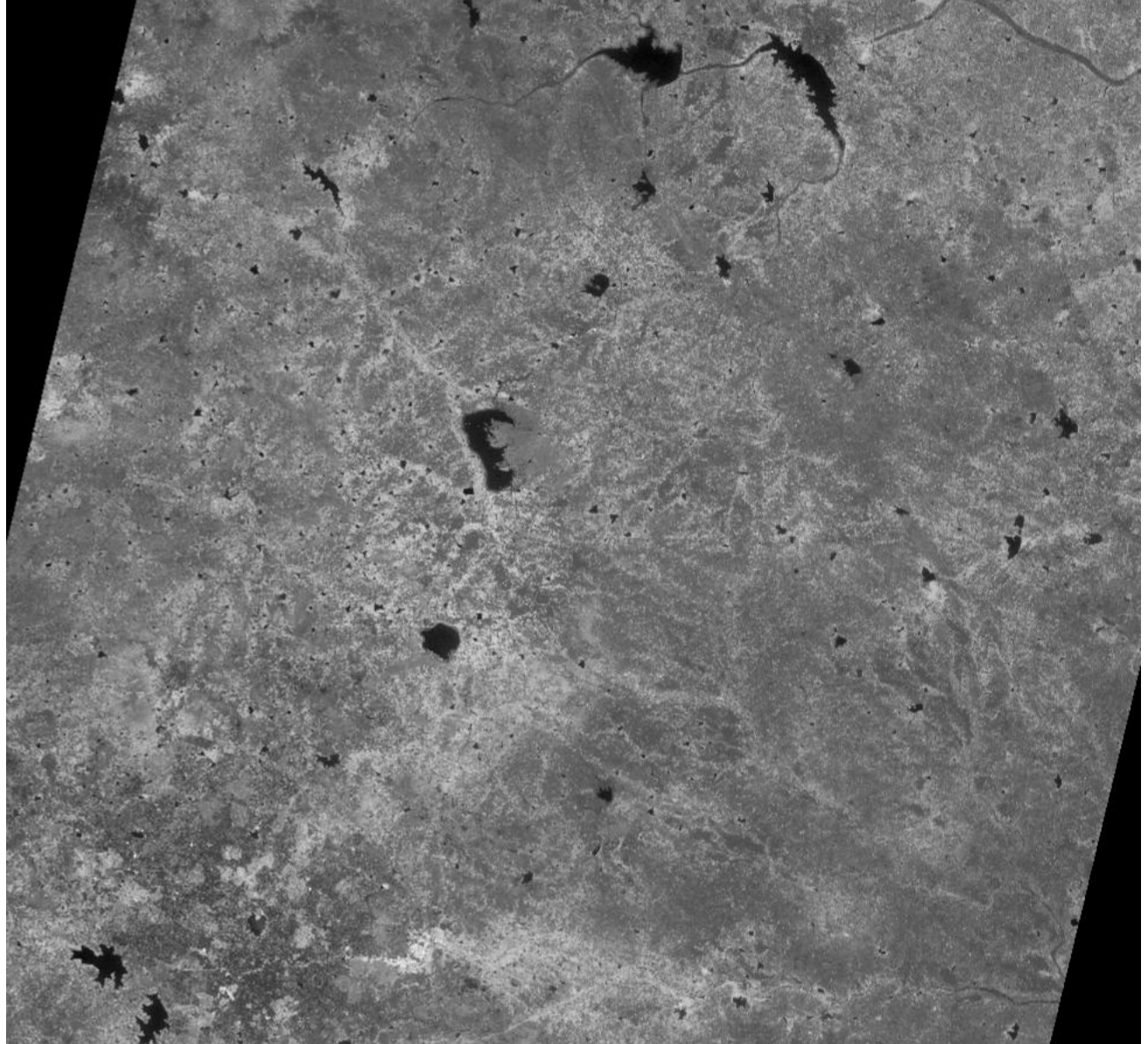


NDVI Expression:



NDVI Output

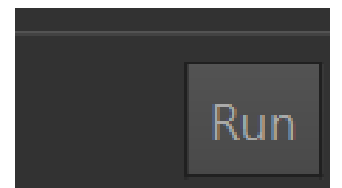
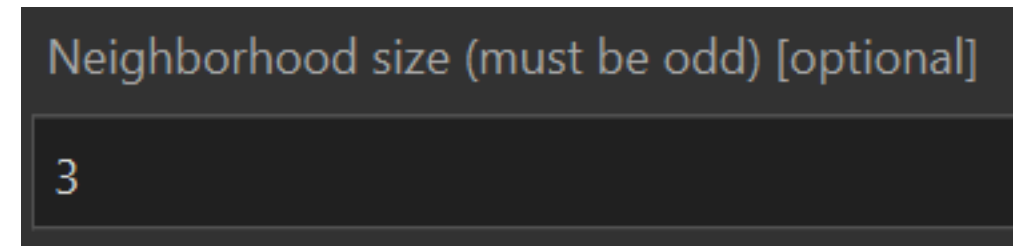
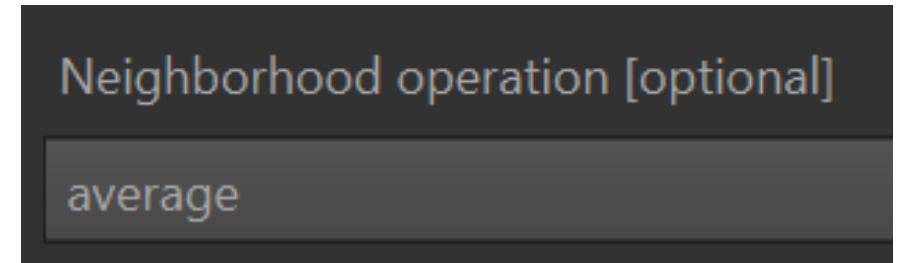
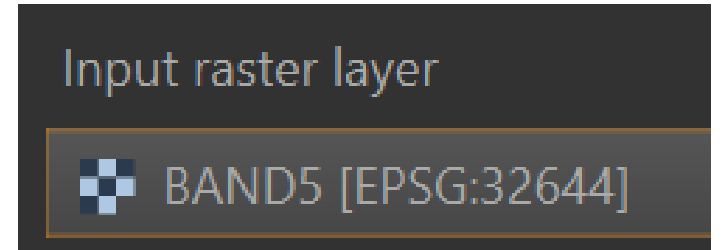
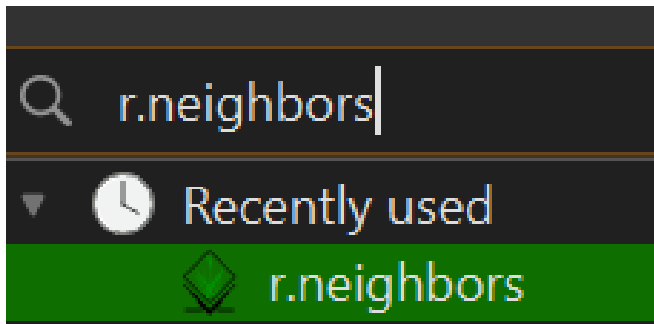
Areas with higher NDVI are brighter,
Whereas with lower NDVI are darker.



Focal: Moving Window

- Select a specific band which shows sharp features over the region and smooth it out with varying neighborhood size 3*3, 5*5

Tool:



Focal: Moving Window

Observe the output

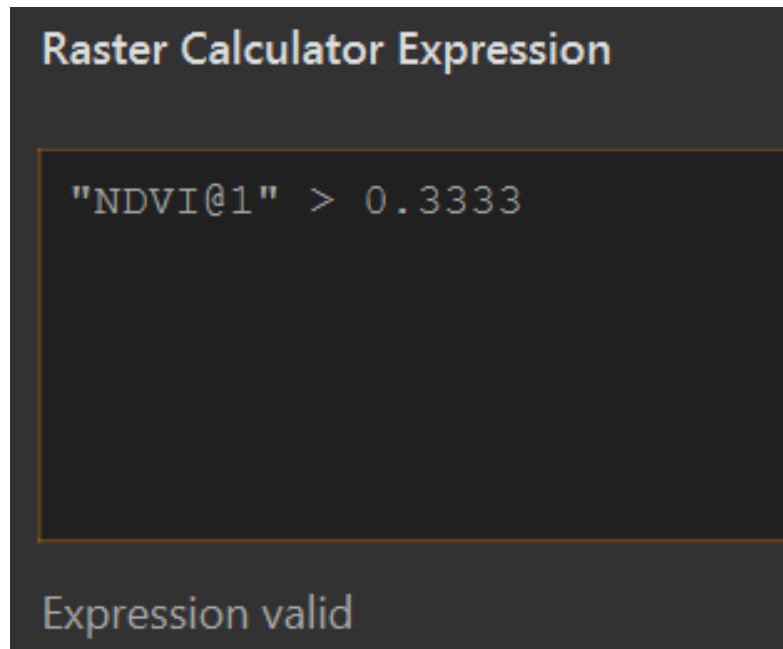
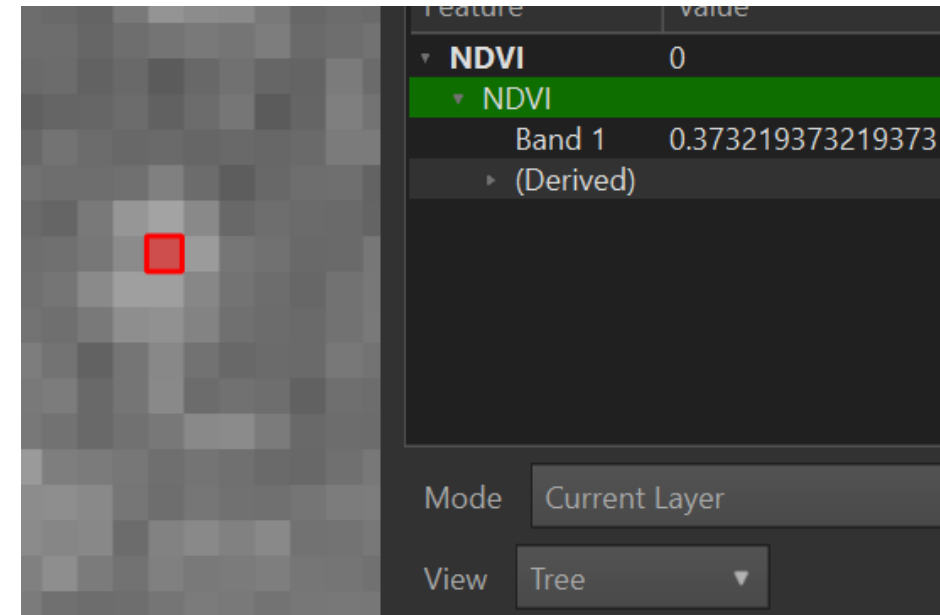
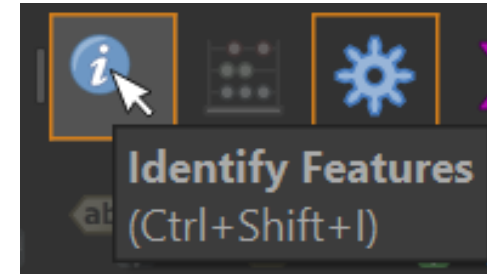
How does the output differ from the original raster?

Thresholding:

Set a threshold value.

Everything greater than threshold = 1

Otherwise = 0



Thresholding Output

Suitable thresholds:

NDVI:

- 0.6 to 1.0 (Healthy, thick plant canopy).
- 0.2 to 0.5 (Shrubs, crops in early stages).
- -0.1 to 0.1 (Barren land, sand).
- Water / Built-up: < 0

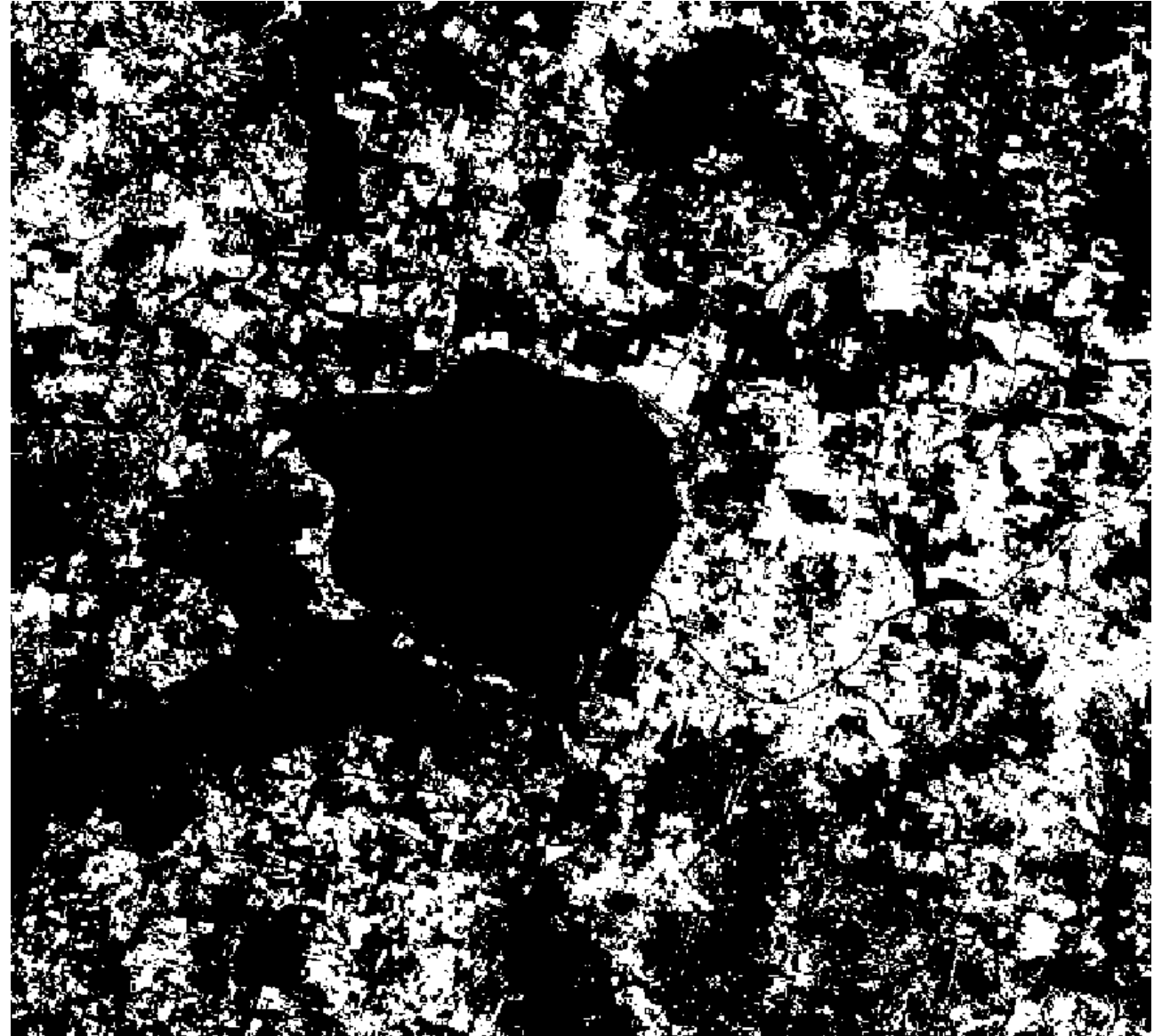
NDWI:

- Water Bodies: > 0.5 (Clear, open water surfaces).
- Flooding / High Moisture: 0.0 to 0.2.
- Drought / Dry Land: < 0 (Non-aqueous surfaces)

NDBI:

- Built-up / Urban Areas: > 0 up to 0.3+
- Vegetation / Water: < 0

Binary raster: Values are either 0 or 1



Reclassification

if structure in Raster Calculator:

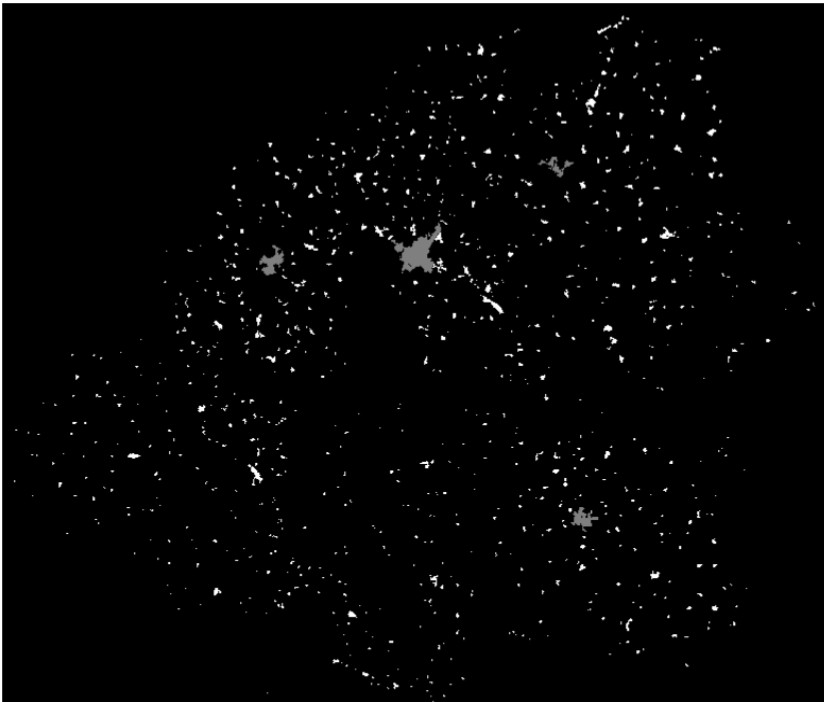
if(condition,value if true, value if false)

Expression for filtering out "Urban"
and "Rural" class

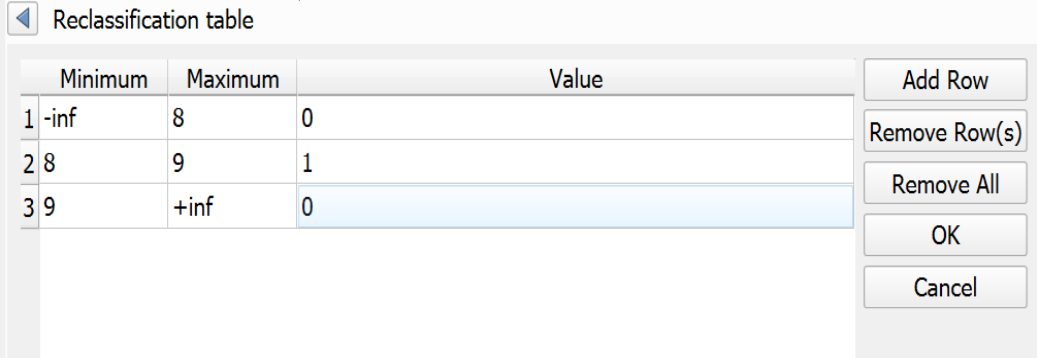
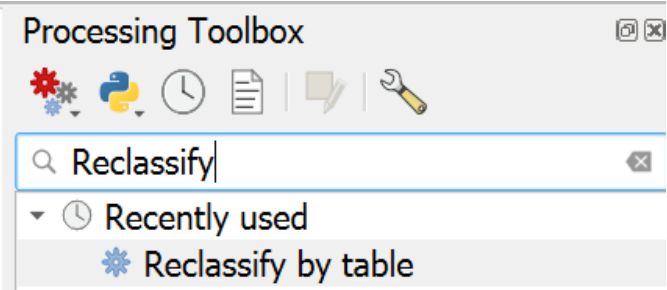
Raster Calculator Expression

```
if ( "LULC_Nizamabad@1" = 1,1,if ( "LULC_Nizamabad@1" = 2,2,0))
```

Output →



Reclassification (Alternative)



Run this

